

Flow A - Detection CI Validation for Wazuh GenAI Content

GitHub PR-triggered detection-as-code quality gate with XML, Sigma, metadata, replay, audit tables, and Slack evidence.

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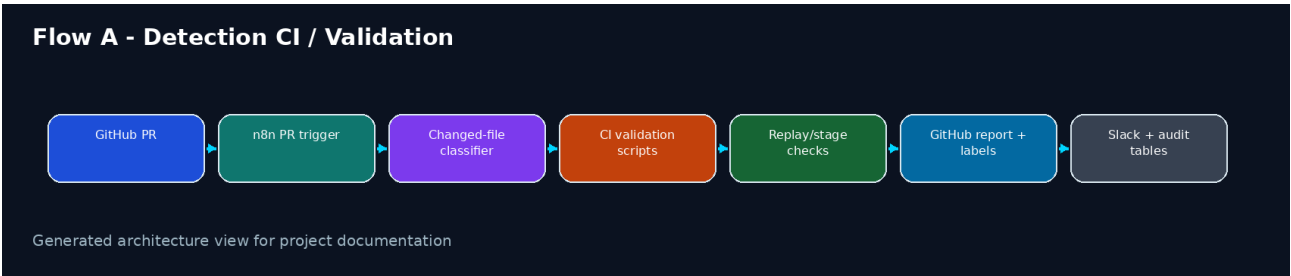
Document Type Workflow project document Flow A	Prototype Focus Detection engineering CI Wazuh GitHub n8n
Project Character Portfolio MVP with implementation, testing, and validation evidence	Design Goal Showcase a practical detection-as-code quality gate before deployment

Why this document matters

- This write-up works both as a portfolio project document and as an implementation-facing prototype note.
- It explains the problem, the workflow design, the SOC value, and the validation evidence that proves the workflow worked in the lab.
- The screenshots are real build/test artifacts collected from n8n, GitHub, Wazuh, Slack, TheHive, and n8n DataTables.

Key outcomes

- Classifies pull request changes and runs only the relevant validation stages.
- Validates Wazuh XML, companion Sigma, metadata mappings, and replay/stage behavior.
- Publishes a GitHub CI report, applies pass/fail labels, sends Slack, and writes DataTable/dashboard evidence.



Architecture view - Flow A turns a GitHub PR into validated detection engineering evidence.

1. Project Summary

Flow A is the detection engineering quality gate for the prototype. It validates detection content before anything is allowed to reach deployment. Instead of relying on manual review alone, it turns each detection-content pull request into a structured CI run with validation stages, GitHub feedback, Slack notification, and audit records.

2. Problem Statement

- Detection content can break Wazuh if XML syntax, decoder logic, or test fixtures are changed without validation.
- Manual review alone is slow and inconsistent when rules, metadata, schemas, and expected outcomes evolve together.
- A SOC detection pipeline needs repeatable evidence before a deployment workflow can trust the content.

3. Objective and Solution Design

- Trigger from GitHub pull request events such as opened, reopened, and synchronized commits.
- Classify changed files into Wazuh rules, Wazuh decoders, Sigma content, test events, expected outputs, and metadata/schema files.
- Run validation only when detection-relevant content changes; skip docs-only PRs with a clear GitHub comment.
- Produce a recruiter-friendly and operations-friendly evidence trail in GitHub, Slack, n8n DataTables, and the SOC dashboard table.

Primary trigger	GitHub pull_request events
Main orchestration	n8n workflow with HTTP, Code, Execute Command, GitHub, Slack, and DataTable nodes
Validation stages	Workspace checkout, Wazuh XML, Sigma, metadata/mapping, staging validation, replay harness
Operational outputs	GitHub comment, PR label, Slack notification, DataTables, SOC dashboard metrics
SOC value	Improves detection quality, reduces broken deployments, and creates a repeatable approval signal for Flow B

4. Implementation Evidence



Figure A1 - Full Flow A n8n workflow canvas showing PR trigger, validation stages, reporting, DataTables, and dashboard posting.

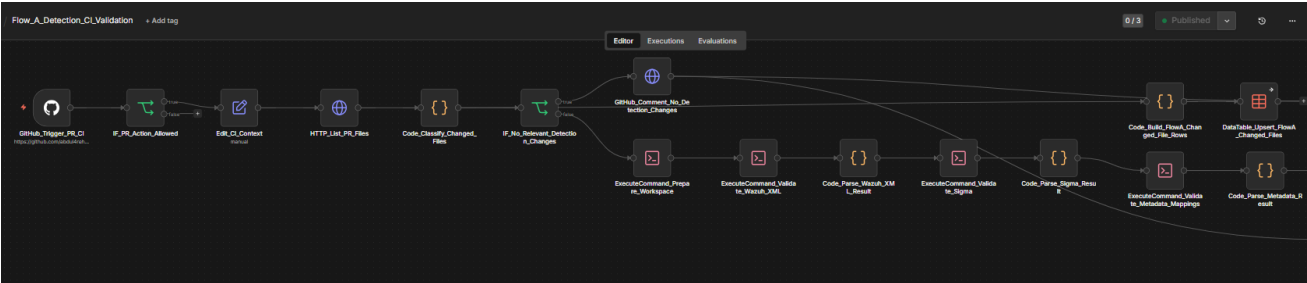


Figure A2 - Flow A left-side zoom: GitHub trigger, PR action filter, context extraction, file listing, and classifier.

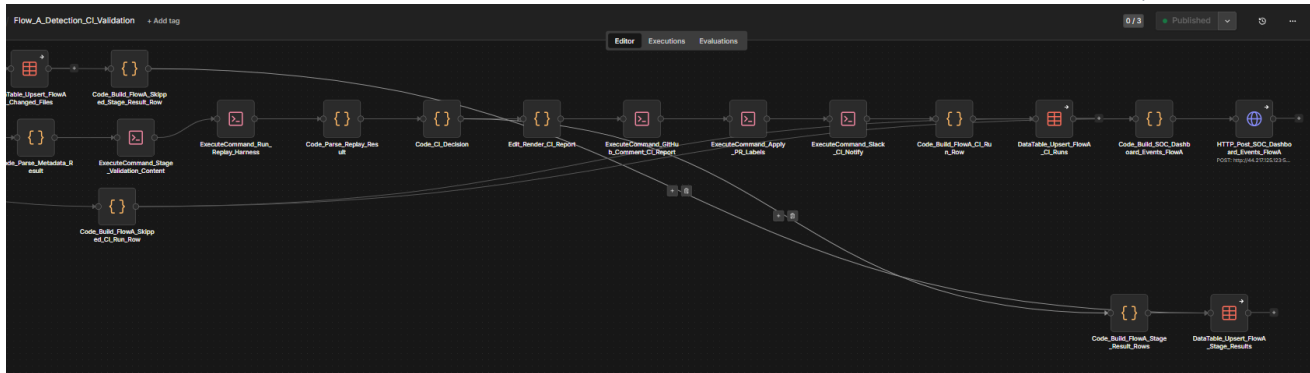


Figure A3 - Flow A right-side zoom: CI decision, reporting, audit DataTables, and SOC dashboard event publishing.

```

GNU nano 7.2 .env.ci
# ----- Flow A deploy settings -----
VALIDATION_WAZUH_MODE=remote_ssh
VALIDATION_WAZUH_HOST=#####
VALIDATION_WAZUH_USER=root
VALIDATION_WAZUH_PORT=22
VALIDATION_SSH_KEY="/opt/detection-ci/wazuh-genai-ci/.runtime-ssh/n8n_to_wazuh_ed25519"
VALIDATION_WAZUH_RULES_DIR=/var/ossec/etc/rules
VALIDATION_WAZUH_DECODERS_DIR=/var/ossec/etc/decoders
VALIDATION_WAZUH_STAGING_DIR=/var/ossec/tmp/detection-ci
VALIDATION_WAZUH_RESTART_CMD="systemctl restart wazuh-manager"

GITHUB_TOKEN="#####"
SLACK_WEBHOOK_URL="#####"

WAZUH_API_URL="https://####:55000"
WAZUH_API_USERNAME="wazuh"
WAZUH_API_PASSWORD="#####"
WAZUH_API_VERIFY_TLS=false
WAZUH_LOGTEST_LOG_FORMAT=syslog
WAZUH_LOGTEST_LOCATION="master->/var/log/syslog"
WAZUH_API_TIMEOUT=30

WAZUH_CUSTOM_RULE_MIN=100000
WAZUH_CUSTOM_RULE_MAX=120000
WAZUH_RULE_ID_ENFORCE=true
  
```

Figure A4 - n8n/CI environment and repository context used for command execution.

```

root@n8n:/opt/detection-ci/wazuh-genai-ci# ls -lh scripts/ci
total 48K
drwxr-xr-x 2 ubuntu ubuntu 4.0K Apr 22 10:18 __pycache__
-rwxr-xr-x 1 ubuntu ubuntu 8.5K Apr 22 10:16 run_replay_harness.py
-rwxr-xr-x 1 ubuntu ubuntu 4.2K Apr 22 10:16 stage_validation_wazuh.sh
-rwxr-xr-x 1 ubuntu ubuntu 4.8K Apr 22 10:16 validate_metadata.py
-rwxr-xr-x 1 ubuntu ubuntu 4.0K Apr 22 10:14 validate_sigma.py
-rwxr-xr-x 1 ubuntu ubuntu 4.6K Apr 22 10:14 validate_wazuh_xml.py
drwxr-xr-x 5 ubuntu ubuntu 4.0K Apr 22 10:17 venv
  
```

Figure A5 - CI scripts present in the repository, including XML, Sigma, metadata, staging, and replay validation helpers.

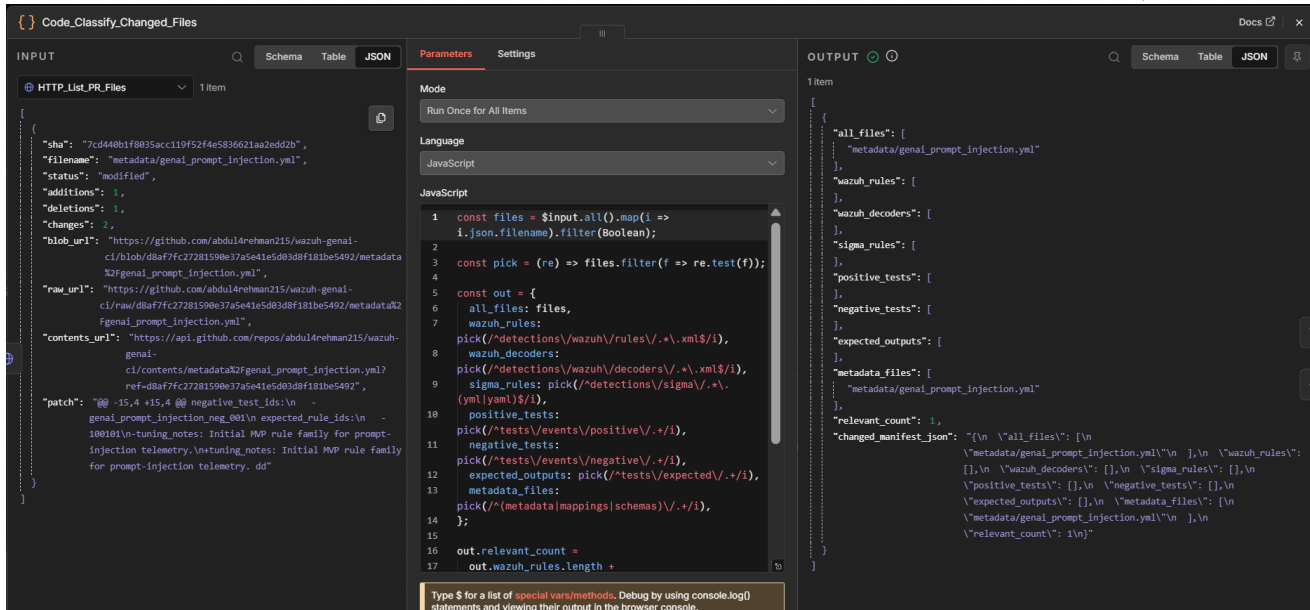


Figure A6 - Code_Classify_Changed_Files output proving metadata-only PR classification for the full CI path.

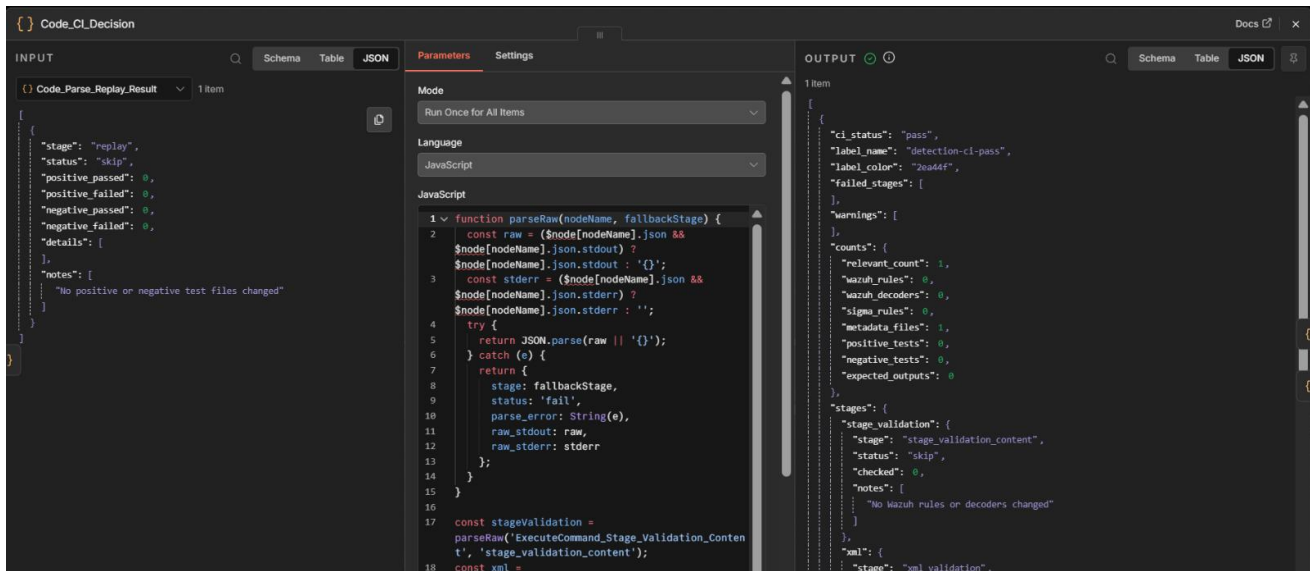


Figure A7 - Code_CI_Decision output showing pass result, no failed stages, and per-stage status details.

5. Validation and Testing Evidence

Test A1 - Docs-only skip path

- A docs-only PR was opened to prove the relevance filter prevents unnecessary validation runs.
- Flow A posted a clear GitHub skip comment and still wrote audit evidence for the changed file and CI run state.

test(flow-a): docs-only skip path #26 [Open](#) [abdul4rehman215](#) wants to merge 1 commit into `main` from `flowa-test-docs-skip-1777128187`

Conversation 0 Commits 1 Checks 0 Files changed 1 +1

abdul4rehman215 commented 2 minutes ago Owner ...

No description provided.

test(flow-a): docs-only skip path 712e53d

abdul4rehman215 commented 2 minutes ago Owner Author ...

Flow A CI skipped.

No detection content changed in this PR.

Checked paths:

- detections/wazuh/rules
- detections/wazuh/decoders
- detections/sigma
- tests/events
- tests/expected
- metadata / mappings / schemas

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications [Customize](#)

[Unsubscribe](#)

Figure A8 - GitHub PR #26 docs-only test showing Flow A CI skipped comment.

GitHub Comment No Detection Changes

Method: POST

URL: `https://api.github.com/repos/abdul4rehman215/wazuh-genei-ci/issues/26/comments`

Authentication: GitHub account

Credential Type: GitHub API

Send Query Parameters: ☒

Send Headers: ☒

Specify Headers: ☐

Headers:

Name	Value
Accept	application/vnd.github+json

Send Body: ☒

Body Content Type: JSON

Specify Body: ☐

OUTPUT

```

{
  "url": "https://api.github.com/repos/abdul4rehman215/wazuh-genei-ci/issues/comments/433886028",
  "html_url": "https://github.com/abdul4rehman215/wazuh-genei-ci/pull/261/discussion/433886028",
  "issue_url": "https://api.github.com/repos/abdul4rehman215/wazuh-genei-ci/issues/26",
  "id": 433886028,
  "node_id": "IC_h4d0337u8AAABkxurJA",
  "user": {
    "login": "abdul4rehman215",
    "id": 101040011,
    "avatar_url": "https://avatars.githubusercontent.com/u/101040011?v=4",
    "gravatar_id": "",
    "url": "https://api.github.com/users/abdul4rehman215",
    "html_url": "https://github.com/abdul4rehman215",
    "followers_url": "https://api.github.com/users/abdul4rehman215/followers",
    "following_url": "https://api.github.com/users/abdul4rehman215/following{/other_user}",
    "gists_url": "https://api.github.com/users/abdul4rehman215/gists{/gist_id}",
    "starred_url": "https://api.github.com/users/abdul4rehman215/starred{/owner}{/repo}",
    "subscriptions_url": "https://api.github.com/users/abdul4rehman215/subscriptions",
    "organizations_url": "https://api.github.com/users/abdul4rehman215/orgs",
    "repos_url": "https://api.github.com/users/abdul4rehman215/repos",
    "events_url": "https://api.github.com/users/abdul4rehman215/events{/privacy}",
    "received_events_url": "https://api.github.com/users/abdul4rehman215/received_events",
    "type": "user",
    "site_admin": false
  },
  "created_at": "2024-04-25T14:43:42Z",
  "updated_at": "2024-04-25T14:43:42Z",
  "author_association": "OWNER",
  "body": "Flow A CI skipped, no detection content changed in this PR. I've checked paths: detections/wazuh/rules, detections/wazuh/decoders, detections/sigma, tests/events, tests/expected, metadata / mappings / schemas.",
  "reactions": {
    "url": "https://api.github.com/repos/abdul4rehman215/wazuh-genei-ci/issues/comments/433886028/reactions",
    "total_count": 0,
    "+1": 0,
    "-1": 0,
    "laugh": 0,
    "hooray": 0,
    "confused": 0,
    "heart": 0,
    "rocket": 0,
    "eyes": 0
  },
  "performed_via_github_app": null,
  "state": null
}

```

Figure A9 - GitHub comment node output confirming the skip report was posted.

A Personal / Data tables / flow_a_ci_runs								
☐	id	T run_id	ts	T owner	T repo	@ pr_number	T head_sha	T actor
☐	1	wazuh-genei-ci-pr-26-712e53d2933ba	2024-04-25T20:13:48.282+05:30	abdul4rehman215	wazuh-genei-ci	26	712e53d2933ba299e07688af464fbc2	abdul4rehman215
☐	2	wazuh-genei-ci-pr-27-d8af7c2728159	2024-04-25T21:18:27.549+05:30	abdul4rehman215	wazuh-genei-ci	27	d8af7c27281590e37a5e1e5d03d8f8	abdul4rehman215

Figure A10 - flow_a_ci_runs table showing a skip row for PR #26 and pass row for PR #27.

Data tables / [flow_a_ci_changed_files](#) :

T run_id	T path	T artifact_type	📅 createdAt	📅 updatedAt
wazuh-genai-ci-pr-26-712e53d2933ba	docs/evidence/flowa-skip-test.md	non_detection_or_other	2026-04-25T20:13:47.720+05:30	2026-04-25T20:13:47.720+05:30
wazuh-genai-ci-pr-27-d8af7fc2728159	metadata/genai_prompt_injection.yml	metadata	2026-04-25T21:18:24.127+05:30	2026-04-25T21:18:24.127+05:30

Figure A11 - flow_a_ci_changed_files table showing docs-only and metadata file classification.

Test A2 - Metadata full CI pass path

- A metadata change was used to trigger the full CI path without unnecessarily deploying Wazuh content.
- The workflow validated metadata successfully, generated a CI report, applied the detection-ci-pass label, and notified Slack.

Update genai_prompt_injection.yml #27

Open

abdul4rehman215 wants to merge 1 commit into main from abdul4rehman215-patch-23

Conversation 0

Commits 1

Checks 0

Files changed 1

abdul4rehman215 commented 1 minute ago

No description provided.

Update genai_prompt_injection.yml

Verifiedd8af7Fc

abdul4rehman215 commented 1 minute ago

OwnerAuthor

Flow A Detection CI Report

Result: PASS
PR: #27
Actor: abdul4rehman215
Head SHA: d8af7Fc

Changed Content Counts

- Relevant changes: 1
- Wazuh rules: 0
- Wazuh decoders: 0
- Sigma rules: 0
- Metadata files: 1
- Positive tests: 0
- Negative tests: 0
- Expected outputs: 0

Stage Results

- Stage validation content: skip
- Wazuh XML validation: skip
- Sigma validation: skip
- Metadata validation: pass
- Replay harness: skip

Failed Stages

- none

Warnings

- none

Figure A12 - GitHub PR #27 Flow A Detection CI Report showing PASS and stage results.

Flow A Detection CI Validation | Portfolio prototype documentation | [abdul4rehman215](#)

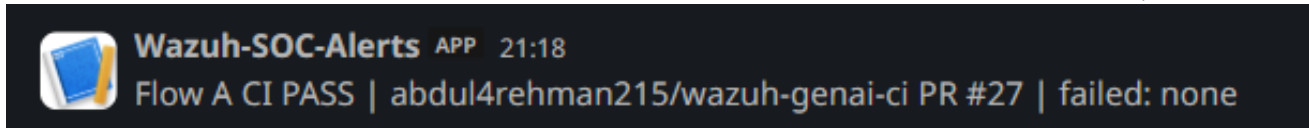


Figure A13 - Slack notification showing Flow A CI PASS.

T_run_id	ts	T_repo	#_pr_number	T_head_sha	T_stage_name	T_stage_status	#_checked	T_passed_files
wazuh-genai-ci-pr-26-712e53d2933ba	2026-04-25T20:13:48.156+05:30	wazuh-genai-ci	26	712e53d2933ba296e27688af464fca2	relevance_filter	skip	0	0
wazuh-genai-ci-pr-27-d8af76c2728159	2026-04-25T21:16:27.763+05:30	wazuh-genai-ci	27	d8af76c27281590e37a5e4fe503d8f8	stage_validation_content	skip	0	0
wazuh-genai-ci-pr-27-d8af76c2728159	2026-04-25T21:16:27.763+05:30	wazuh-genai-ci	27	d8af76c27281590e37a5e4fe503d8f8	wazuh_xml_validation	skip	0	0
wazuh-genai-ci-pr-27-d8af76c2728159	2026-04-25T21:16:27.763+05:30	wazuh-genai-ci	27	d8af76c27281590e37a5e4fe503d8f8	sigma_validation	skip	0	0
wazuh-genai-ci-pr-27-d8af76c2728159	2026-04-25T21:16:27.763+05:30	wazuh-genai-ci	27	d8af76c27281590e37a5e4fe503d8f8	metadata_validation	pass	1	1
wazuh-genai-ci-pr-27-d8af76c2728159	2026-04-25T21:16:27.763+05:30	wazuh-genai-ci	27	d8af76c27281590e37a5e4fe503d8f8	replay_harness	skip	0	0

Figure A14 - flow_a_ci_stage_results table showing each validation stage and pass/skip status.

#_metadata_files	#_positive_tests	#_negative_tests	#_expected_outputs	T_pr_url	T_label_name	T_slack_status	T_comment_stage
0	0	0	0	https://github.com/abdul4rehman215/wazuh-genai-ci/pull/26	flow-a-skipped-no-detection-changes	not_sent	github_comment_no_detection_change
1	0	0	0	https://github.com/abdul4rehman215/wazuh-genai-ci/pull/27	detection-ci-pass	Empty	github_comment_report

Figure A15 - flow_a_ci_runs continuation columns with PR URL, label, Slack, and comment stage values.

6. SOC Value and Impact

- Improves detection engineering consistency by making validation repeatable and evidence-backed.
- Reduces deployment risk because Flow B can use detection-ci-pass as a trustworthy gate condition.
- Reduces analyst and engineer rework by catching invalid content before it reaches the Wazuh manager.
- Creates structured data for later dashboarding and audit review.

7. Skills Demonstrated

- Detection-as-code pipeline design
- n8n orchestration
- GitHub API integration
- Wazuh validation automation
- JSON parsing and decision logic
- Slack reporting
- DataTable audit modeling

8. Prototype Limitations and Future Enhancements

- Prototype uses a lab repository and self-hosted n8n instance rather than enterprise CI infrastructure.
- Future expansion could add branch protection integration, signed validation artifacts, richer test coverage, and pull request status checks.
- Additional detection families can be added by extending metadata rules and replay test fixtures.

9. Conclusion

Flow A establishes the quality foundation for the complete MVP. It proves that GenAI Wazuh detection content can be validated, reported, and audited before a controlled deployment is allowed.